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Number of packets in transit as a function of source load and routing

Anna T. Lawniczak*, Shengkun Xie

Department of Mathematics and Statistics, University of Guelph, Guelph, Ontario, N1G 2W1, Canada

Abstract

We study how network performance in delivering packets to their destinations is affected by routing algorithm coupled with volume of incoming traffic. We focus our study on the *number of packets in transit (NPT)* that is an aggregate measure of a network quality of service (QoS) performance. The *NPT* network performance indicator measures directly the number of packets in the network on their routes to their destinations. We carry out our study using a time-discrete simulation model that is an abstraction of the Network Layer of the ISO OSI Reference Model. This model focuses on packets and their routing. We consider a static routing and two different types of dynamic routings and different volumes of incoming traffic in the network free flow state. Our study shows that the efficiency of performance of a routing measured as an average value of the *NPT* time series and as a variability of this series, changes with the volume of incoming traffic among the considered routings. Thus, depending on the volume of incoming traffic it is preferable to use one type of routing over the other ones if the objective is to maintain the lowest number of packets in transit and their variability, i.e. the highest QoS network performance.

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Keywords: Packet switching network; quality of service; efficiency; stability; number of packets in transit; free flow

1. Introduction

The most common technology of current data communication networks (e.g., the Internet, wireless networks, ad-hoc networks, etc.) is Packet Switching Network (PSN) [1], [2]. Data communication networks are complex systems, performance of which depends on many factors and on coupling among these factors, i.e. network connection topology, routing, volume of incoming packet traffic, and its type, etc. To better understand how network performance and packet traffic dynamics depends on network factors and on their couplings often it is useful to consider simplified models of PSNs, e.g. [3]–[9]. We limit our study of PSNs to the Network Layer of the International Standard Organization Open Systems Interconnect (ISO OSI) Reference Model [1], [2]. In our study we use an abstraction of the Network Layer developed in [5], [6] and its time-discrete simulation model [10], [11]. This abstraction focuses on packets and their routing. The goal of our investigation is to study how routing coupled with volume of incoming traffic (source load value) impacts network quality of service (QoS) performance in

* Corresponding author. Tel.: 001-519-824-4120 ext 53287; fax: 001-519-837-0221.

E-mail address: alawnicz@uoguelph.ca

delivering packets to their destinations. We focus our study on the *number of packets in transit (NPT)* that is an aggregate measure of the network performance. This network performance indicator measures directly how many packets are in the network on their routes to their destinations. Thus, it provides direct information on how the coupling of a source load value with a given type of routing affects network efficiency and stability, which is measured by an average value of the *NPT* time series and its variability. We consider a static routing and two different types of dynamic routings and source load values in the network free flow state. Our study shows that depending on a source load value one or another type of routing is more efficient. For example, among the considered routings, for small source load values the *NPT* values are the lowest for the static routing. However, they are the highest for the source load values approaching the phase transition point from free flow to the congested network state, i.e. the *critical point*. For these source load values the dynamic routing, which we call *QSPO*, based on routing decisions taking under consideration a number of hops that a packet must perform from its current location to its destination and the information how congested the routers are, performs the best. Also, our study shows that variability in number of packets in transit depends on the coupling of a source load value with a given type of routing. For example, among the considered routings, for small source load values the variability in number of packets in transit is the lowest when the static routing is used. However, when the source load values approach the *critical point* the variability in number of packets in transit is the highest for the static routing and the lowest for the dynamic routing *QSPO*. This means that for small source load values the static routing provides the highest QoS. While, for high source load values the static routing provides the lowest QoS and the dynamic routing *QSPO* provides the highest QoS.

Our study shows, that among the considered routings the network efficiency and stability in delivering packets to their destinations, i.e. the network QoS, depend on a source load value. Thus, for a given source load value it may be preferable to use one type of routing than the other ones if the objective is to maintain the lowest number of packets in transit and their variability, i.e. the highest QoS of network performance.

2. Description of Packet Switching Network Model

We briefly describe the PSN model, developed in [5], [6], and its C++ simulator, called *Netzwerk-1* [10], [11] that we use in our study. The PSN model is an abstraction of the ISO OSI Network Layer Reference Model. The PSN model focuses on packets and their routings. It is scalable, distributed in space, and time discrete. It avoids the overhead of protocol details present in many PSN simulators designed with different aims in mind.

In the PSN model each node performs the functions of *host* and *router* and maintains one incoming and one outgoing queue which is of unlimited length and operates according to a first-in, first-out policy. At each node, independently of the other nodes, packets are created randomly with probability λ called *source load*. In the PSN model all messages are restricted to one packet carrying only the following information: time of creation, destination address, and number of hops taken.

The PSN model connection topology is represented by a weighted directed multigraph L where each node corresponds to a vertex and each communication link is represented by a pair of parallel edges oriented in opposite directions. To each edge is assigned a cost of packet transmission. For a given PSN model set-up all edge costs are computed using the same type of *edge cost function (ecf)* that is either the *ecf* called *ONE (ONE)*, or *QueueSize (QS)*, or *QueueSizePlusOne (QSPO)*. The *ecf ONE* assigns a value of “one” to all edges in the lattice L . The *ecf QS* assigns to each edge in the lattice L a value equal to the length of the outgoing queue at the node from which the edge originates. The *ecf QSPO* assigns a value that is the sum of a constant “one” plus the length of the outgoing queue at the node from which the edge originates. The *ecf ONE* edge cost does not change during a simulation run, thus this results in a static routing. Since the routing decisions made using the *ecf QS* or *QSPO* rely on the current state of the network simulation this implies *adaptive* or *dynamic routing*. In the PSN model, each packet is transmitted via routers from its source to its destination according to the routing decisions made independently at each router and based on a *minimum least-cost criterion*. During a simulation of the PSN model using dynamic routing packets have the ability to avoid congested nodes, they do not have this ability when static routing is used instead.

In the PSN model time is discrete and we observe the network state at the discrete times $k = 0, 1, 2, \dots, T$, where T is the final simulation time. At time $k = 0$, the setup of the PSN model is initialized with empty queues and the routing tables are computed. The time-discrete, synchronous and spatially distributed PSN model algorithm consists

of the sequence of five operations advancing the simulation time from k to $k + 1$. These operations are: (1) *Update routing tables*, (2) *Create and route packets*, (3) *Process incoming queue*, (4) *Evaluate network state*, (5) *Update simulation time*. The detailed description of this algorithm is provided in [5], [6].

A PSN model setup is defined by a selection of: a type of network connection topology, a type of *ecf*, a type of routing table and its update algorithm, a value of source load, seeds of two pseudo-random number generators and a final simulation time T . The first pseudo-random number generator provides the sequence of numbers required for packets generation and routing. The second one is used for specifying the network connection topology. The details of PSN model setup are provided in [6].

3. PSN Model Simulation Experiments and Network Performance Indicators

In this paper we analyze simulation data for the PSN model setups with a network connection topology that is isomorphic to $L^p_{\square}(16)$ (i.e., a two-dimensional periodic square lattice with 16 nodes in the horizontal and vertical directions), full-table routing and distributed routing table update and we use the default value for the second pseudo-random number generator. The incoming traffic is generated, at the network nodes, by Bernoulli random variables with expected value λ , i.e. source load value. In our simulation experiments we vary the values of the following setup variables: *ecf*, source load and seed of the first pseudo-random number generator. Some times we call these setup variables as factors or treatments. We use, respectively, the following conventions $L^p_{\square}(16, ecf)$ and $L^p_{\square}(16, ecf, \lambda)$, where *ecf* = *ONE*, or *QS*, or *QSPO*, when we want to specify with what type of *ecf* and additionally with what value of source load λ the PSN model is setup.

In the PSN model, for each family of network setups, which differ only in the value of the source load λ , values of $\lambda_{\text{sub-c}}$ for which packet traffic is congestion-free are called *sub-critical source loads*, while values $\lambda_{\text{sup-c}}$ for which traffic is congested are called *super-critical source loads*. The *critical source load* λ_c is the largest sub-critical source load. Thus, λ_c is an important network performance indicator because it is the phase transition point from free flow to congested state of a network. Details about how we estimate the critical source load are provided in [6].

For the PSN model setups considered here the estimated *critical source load* (CSL) values are, respectively, $\lambda_c = 0.115$ for $L^p_{\square}(16, \text{ONE})$, $\lambda_c = 0.120$ for $L^p_{\square}(16, \text{QS})$ and $\lambda_c = 0.120$ for $L^p_{\square}(16, \text{QSPO})$ [12], [13].

Our study focuses on the free flow state for each PSN model setup. We analyze the performance of PSN model setups $L^p_{\square}(16, ecf, \lambda)$, for *ecf* = *ONE*, *QS*, *QSPO*, sub-critical source load values $\lambda = 0.005, 0.020, 0.030, 0.040, 0.050, 0.060, 0.070, 0.080, 0.090, 0.095, 0.100, 0.105$ and 0.110 , 24 different seed values l , where $l = 1, \dots, 24$, of the first pseudo-random number generator. Thus, we consider source load values from very small ones to those close to the phase transition point and investigate how an experimental (simulation) setup of PSN model (i.e., treatment) affects the behaviour of the response variable *number of packets in transit* [6]. We call the considered set of λ values *FreeFlow* set. We carry out our investigations for the final simulation time $T = 8000$. Even though the final simulation time is $T = 8000$ only the data from $k = 2001$ is accounted for in order to remove the initial transient effects caused by the setups of the PSN model always with empty queues. Thus, notice, in all the presented graphs time axis scale goes always from 0 to 6000 to account for the discarded data.

Number of packets in transit (NPT), i.e. $N(k)$ time series, is given by the total number of packets in the network at time k , i.e. by the sum over all network nodes of the number of packets in each out-going queue at time k . The NPT time series is an important time dependent aggregate measure of network performance providing information on how many packets are in the network on their routes to their destinations at time k for a given PSN model setup.

Fig. 1 shows the graphs of $N_l(k)$ time series for $k = 2001$ to 8000 for PSN model setup $L^p_{\square}(16, ecf, \lambda)$ with *ecf* *ONE*, *QS* and *QSPO* (left, middle and right column, respectively) and $\lambda = 0.040, 0.080$ and 0.105 (top, middle and bottom row, respectively) for $l = 1, \dots, 24$ considered seed values of the first pseudo-random number generator. We observe that

1. each $N_l(k)$ time series fluctuates around a constant mean depending on *ecf* type and source load value;
2. for both $\lambda = 0.040$ and $\lambda = 0.080$ when *ecf* *QS* is used the respective mean values of the NPT series are much higher than the mean values when *ecf* *ONE* or *ecf* *QSPO* is used;
3. for $\lambda = 0.105$ when *ecf* *ONE* and *ecf* *QS* is used, respectively, the corresponding mean values of the NPT series appear to be very similar ones and they are both higher than the mean value when *ecf* *QSPO* is employed;

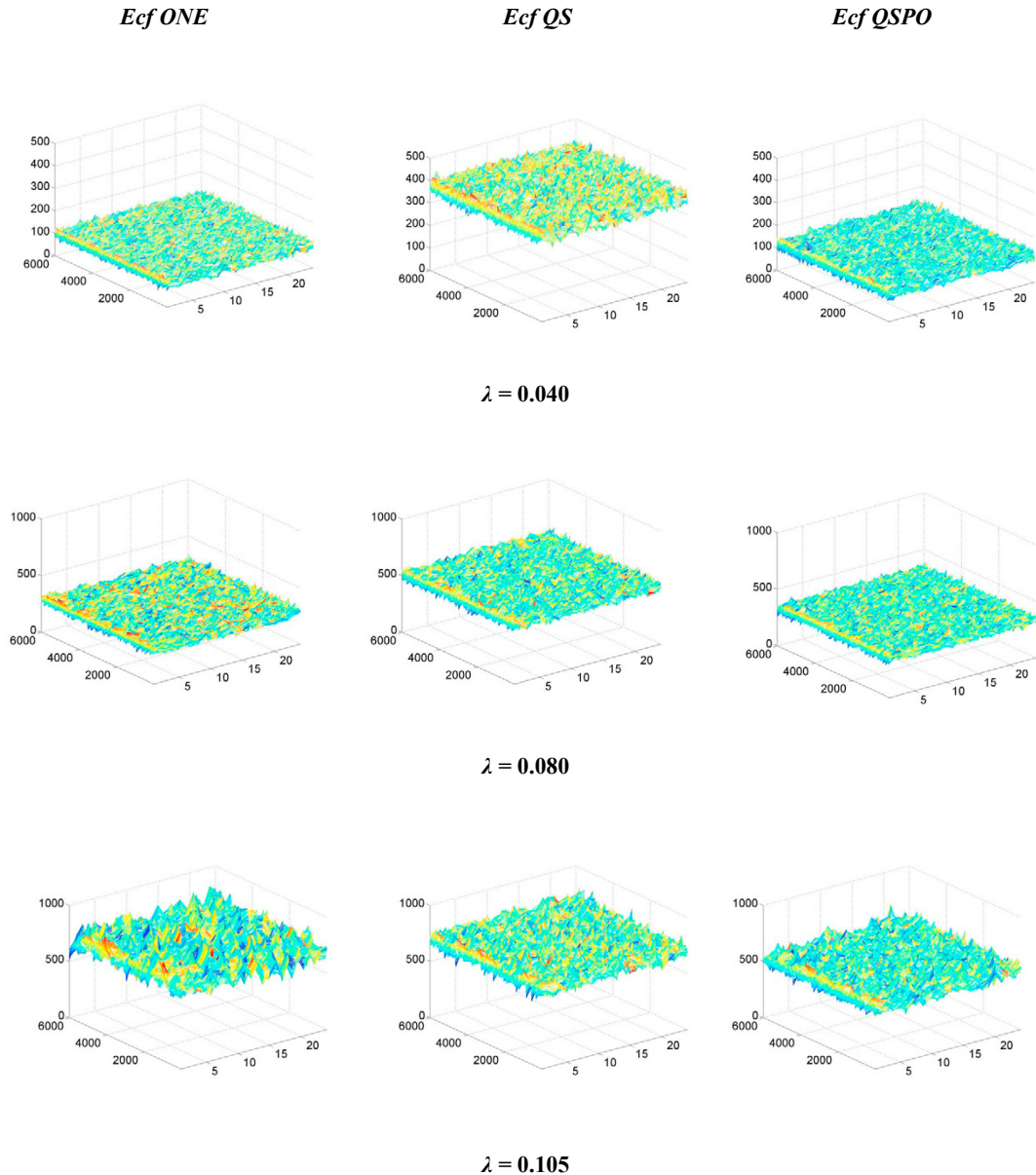


Fig. 1 Graphs of *NPT* time series $N_l(k)$ for PSN model setup $L^p_{\square}(16, ecf, \lambda)$ with $ecf = ONE, QS, QSPO$ (left, middle, right column, respectively) and $\lambda = 0.040, 0.080, 0.105$ (top, middle, bottom row, respectively) corresponding to seed values l , where $l = 1, \dots, 24$, of the first pseudo-random generator. The values of the seeds are plotted on the x -axis, time k on the y -axis and the $N_l(k)$ values on the z -axis.

4. with the increase of source load values the order of the magnitudes of the *ecf* dependent mean values changes;
5. with the increase of the source load values the magnitudes of fluctuations of $N_l(k)$ time series increase.

The plots on Fig. 1 provide some qualitative insight into how the packet traffic dynamics and network QoS performance change with the increase of source load values for the considered *ecfs*. To better understand these changes and to quantify them we study statistical characteristics/measures of *NPT* time series. Namely, source load dependent mean value functions $\hat{N}(ecf, \lambda)$ and source load dependent variance functions $\text{Var } N(ecf, \lambda)$.

4. Impact of Source Load and ECF on Number of Packet in Transit

We measure the QoS of a PSN model setup $L^p_{\square}(16, ecf, \lambda)$ by the efficiency with which packets are delivered to their destinations and by the stability in number of packets in transit. We measure a routing efficiency by mean value of the *NPT* time series and we measure the routing stability by variability of these series, in both cases calculated as averages over the length of the considered simulation run time and the number of the simulation runs. The smaller the values of these statistical measures of network performance the better QoS provides a PSN model setup, i.e. the routing is more efficient and stable in delivering the packets to their destinations, as on average there are fewer packets in transit in the network and with smaller fluctuations. We investigate if for sub-critical source load values the QoS in delivering packets to their destinations, i.e. efficiency and stability, changes among various *ecfs* (i.e., routings) when source load values increase.

Fig. 2 displays the graphs of source load dependent mean value functions $\hat{N}(ecf, \lambda)$ in (a), and the graphs of source load dependent variance functions $\text{Var } N(ecf, \lambda)$ in (b), of $N_l(k)$ times series, where $l = 1, \dots, 24$ is, respectively, the seed value l of the first pseudo-random number generator. As mentioned earlier, these statistical measures of network performance are calculated as averages over the number of simulation samples (i.e., 24) and over the length of the considered simulation run time (i.e., 6000) for each PSN model setup $L^p_{\square}(16, ecf, \lambda)$, with *ecf* = *ONE*, *QS*, *QSPO*, respectively, and λ belonging to the *FreeFlow* set.

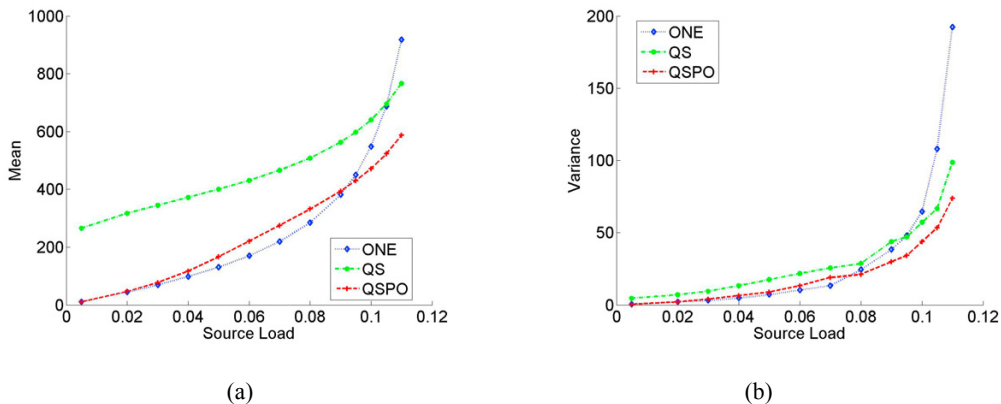


Fig. 2 Graphs of source load dependent mean value functions $\hat{N}(ecf, \lambda)$ are in (a), and of source load dependent variance functions $\text{Var } N(ecf, \lambda)$ are in (b), of $N_l(k)$ times series where $l = 1, \dots, 24$. These statistical measures of network performance are calculated as averages over the considered simulation time (i.e., 6000) and over the number of simulation samples (i.e., 24) corresponding to different seed values l of the first pseudo-random number generator for each PSN model setup $L^p_{\square}(16, ecf, \lambda)$, with *ecf* = *ONE*, *QS*, *QSPO*, respectively, and λ belonging to the *FreeFlow* set.

On Fig. 2 (a) we observe that

1. the values of all three mean value functions monotonically increase with the increase of λ ;
2. the rates of increase of the mean value functions $\hat{N}(QS, \lambda)$ and $\hat{N}(QSPO, \lambda)$ are similar ones and they appear to be smaller than the rate of increase of the function $\hat{N}(ONE, \lambda)$;
3. for $\lambda < 0.100$ the values of the mean value function $\hat{N}(QS, \lambda)$ are much larger than those of the functions $\hat{N}(ONE, \lambda)$ and $\hat{N}(QSPO, \lambda)$;
4. for source loads $\lambda < 0.095$ the values of the mean value functions $\hat{N}(ONE, \lambda)$ and $\hat{N}(QSPO, \lambda)$ do not differ much, but they differ significantly for $\lambda \geq 0.100$;
5. the order of the magnitudes of the mean value functions $\hat{N}(ecf, \lambda)$ changes with the increase of source load values and in the *FreeFlow* set we can identify 4 different subsets such that within each of these subsets the order stays the same. Namely, (i) $\hat{N}(QS, \lambda)$ is significantly greater than both $\hat{N}(ONE, \lambda)$ and $\hat{N}(QSPO, \lambda)$, and $\hat{N}(ONE, \lambda) \approx \hat{N}(QSPO, \lambda)$, for each $\lambda \leq 0.030$ (approximately); (ii) $\hat{N}(ONE, \lambda) < \hat{N}(QSPO, \lambda) < \hat{N}(QS, \lambda)$ for $0.030 < \lambda \leq 0.090$; (iii) $\hat{N}(QSPO, \lambda) < \hat{N}(ONE, \lambda) < \hat{N}(QS, \lambda)$ for $0.090 < \lambda < 0.105$; (iv) $\hat{N}(QSPO, \lambda) < \hat{N}(QS, \lambda) < \hat{N}(ONE, \lambda)$ for $\lambda \geq 0.105$.

When one focuses only on the mean values of the *NPT* time series, we can conclude that the network efficiency of delivering packets to their destinations is *ecf* and load dependent. The observations stated in (i)-(iv) in the item 5 above imply that for the values of $\lambda \leq 0.090$ the static routing using *ecf ONE* is the most efficient one in delivering packets to their destinations, as the average number of packets in the network on their routes to their destinations is the lowest one. The packets travel along the shortest paths (i.e., the paths with the minimum number of hops) and they do not queue for too long at the routers as there are seldom local congestions along the paths in free flow state of the network. The efficiency of the dynamic routing employing *ecf QSPO* is close to the one using *ecf ONE*, due to the *ONE* component of the *ecf QSPO* that pushes the routing to select the paths with minimum number of hops from packets current locations to their destinations. The dynamic routing using *ecf QS* is the least efficient one as the average number of packets in transit is the highest. The reason for this is that the least cost routing using *ecf QS* always sends the packets from their current nodes to their destinations along the routes which have the smallest sums of queues. This routing tries to avoid congested routers and as a result redistribute the packets rather evenly among the routers. It does not take under consideration through how many routers packets must hop to reach their destinations. These result in packets performing almost random walks as many paths from packets current locations to their destinations may have very similar costs. Thus, packets stay longer in the network as they do not have enough clues how to reach their destinations. This is prevented by the *ONE* component of the *ecf QSPO*. When $0.090 < \lambda < 0.105$ the dynamic routing using *ecf QSPO* becomes the most efficient one, as it takes under consideration not only the number of hops that packets must perform to reach their destinations but also the queue sizes of the routers located along the packets routes. This allows packets to avoid locally congested network routers. As before the routing using *ecf QS*, i.e. the routing that is only router load dependent, is the least efficient one. When $\lambda \geq 0.105$, i.e. for λ close to the critical points λ_c the most efficient routing is the one using *ecf QSPO* and the least efficient one is the one using *ecf ONE*. The reason for this is that the local congestions build up and the dynamic routing using *ecf QSPO* or *ecf QS* has the ability to avoid locally congested network routers. The static routing does not have this ability and packets accumulate further in the already locally congested network routers, as can be seen from Fig. 1 for $\lambda = 0.105$.

On Fig. 2 (b) we observe that

1. the values of all three load dependent variance functions $\text{Var } N(ecf, \lambda)$ are monotonically increasing with the increase of λ . The rate of increase of these functions is similar up to $\lambda \approx 0.095$. Pass this point the function $\text{Var } N(ONE, \lambda)$ has much higher rate of increase than the two other functions;
2. for each $\lambda < 0.090$ the values of the load dependent variance functions $\text{Var } N(ecf, \lambda)$ differ only slightly for all three *ecfs*. However, the values of $\text{Var } N(QS, \lambda)$ are higher than those of $\text{Var } N(ONE, \lambda)$ and $\text{Var } N(QSPO, \lambda)$. This means that the *NPT* data of the PSN model using *ecf QS* has larger variability around its mean than the *NPT* data of the PSN model using either one of the two others *ecfs*.
3. the values of the load dependent variance functions $\text{Var } N(ecf, \lambda)$ diverge rapidly for $\lambda > 0.90$ and these values satisfy the inequalities $\text{Var } N(ONE, \lambda) > \text{Var } N(QS, \lambda) > \text{Var } N(QSPO, \lambda)$. This means that the *NPT* data of the PSN model using *ecf ONE* has the largest variability around its mean and the *NPT* data of the PSN model using *ecf QSPO* has the lowest variability around its mean;

4. for each *ecf* type the rate of increase of its variance function $\text{Var } N(\text{ecf}, \lambda)$ increases rapidly when the source load values λ approach the respective λ_c values, (i.e., for $0.095 \leq \lambda \leq 0.115$), in particular, this is the case for the variance function $\text{Var } N(\text{ONE}, \lambda)$.

From the above we can conclude that the magnitudes' of fluctuations in the *NPT* time series around their respective means are similar ones for all three *ecfs* when $\lambda < 0.090$, implying that the stabilities of the corresponding routings are similar ones. However, the magnitudes' of fluctuations are significantly different when $\lambda > 0.090$. The *NPT* data of the PSN model using *ecf ONE*, i.e. using the static routing, shows the highest variability. Thus, the static routing shows the highest instability among the routings for $\lambda > 0.090$. As the source load values approach the critical points the local congestions build up randomly. Since the static routing does not have the ability to send packets avoiding congested network routers this leads to high random fluctuations in packet traffic over time and over different simulations. The least variability shows the *NPT* data of the PSN model using *ecf QSPO*, i.e. the dynamic routing using the *ecf* combining both the router load dependent and the static cost components. The router load dependent cost component of the *ecf QSPO* pushes the routing to send packets along the routes avoiding locally congested routers and the static cost component of *ecf* pushes the routing to select the routes with minimum number of hops from packets current locations to their destinations. The superposition of these two cost components results in the routing using *ecf QSPO* being the most efficient one (see Fig. 1) and the most stable one (see Fig. 2) when source load values are high. The considered dynamic routings send packets along the routes avoiding congested network routers; they "smooth out" the packet distributions among the routers. This results in higher routing stability as there is less random fluctuation in the packet traffic over time and over different simulations, in particular when $\lambda > 0.090$.

When the least cost dynamic routing bases on the *ecf QS*, i.e. on the *ecf* depending only on the routers' loads, as a response to the queue size changes and rapid routing table updates, poor route selections may happen causing oscillatory packet behaviours. This leads to packets staying unnecessarily long in transit (see Fig. 1) and to the increased fluctuations in their numbers (see Fig. 2). This is why the QoS of a network using the *ecf QS* is worst than the QoS of a network using *ecf QSPO*.

Looking at the plots on Fig. 1 and Fig. 2 we may conclude that the QoS of a network in delivering packets to their destinations depends on the combination of the routing and source load value. Thus, the QoS can be maximized by selecting a routing type depending on an amount of the incoming traffic. Furthermore, the dynamic routing using the *ecf QSPO* provides overall a good QoS in delivering packets to their destinations regardless what is the source load value. Also, the observed increase in values of the variance functions $\text{Var } N(\text{ecf}, \lambda)$, i.e. of fluctuations of *NPT* data, when source loads approach the respective critical points, i.e. phase transition points from free flow to congested network states, is consistent with the phase transition phenomena and observed behaviours of packet traffic in data networks [3], [6]–[9].

5. Conclusions and Future Work

The above discussed source load dependent mean value functions $N(\text{ecf}, \lambda)$ and source load dependent variance functions $\text{Var } N(\text{ecf}, \lambda)$ are statistical aggregate measures of network performance that provide some insight into the number of packets in transit and their variability. Thus, these functions provide insight into how efficiently the routing delivers the packets to their destinations and how it is stable for a given *ecf* type and source load value. Our study shows that the efficiency of performance of a routing, measured as an average value of the *NPT* time series and the stability measured as a variability of these series, change with source load values among the considered *ecfs*. Thus, depending on a source load value, to provide the highest QoS in delivering packets, it may be preferable to use one type of *ecf* over the other ones if the objective is to maintain the lowest number of packets in transit and their variability. The network using the static routing provides the highest QoS in delivering packets to their destinations when the source load values are far from the critical points. However, for source load values close to the critical points the dynamic routings (i.e., the routing using *ecf QS* or *QSPO*) are more efficient ones than the static routing. The dynamic routings have the ability to send packets along the routes avoiding congested network nodes. This allows packets to reach their destinations faster and results in lower number of packets in transit. Also, they distribute the packets more evenly among the network routers. This results in less fluctuation in the packet traffic that is reflected by the lower variability in the *NPT* data. The dynamic routing using the *ecf QSPO* provides

overall a good QoS in delivering packets to their destinations regardless what is the source load value. For the source load values close to the critical points it provides the highest QoS in delivering packets.

Since the considered mean value and variance functions are time averages they ignore time differences among the *NPT* time series. They provide coarse granularity, both spatially and temporally, information about network-wide QoS performance. We plan to refine our the analysis by investigating the time differences among *NPT* time series using functional analysis of variance (FANOVA) model [14] and explore how time variability of *NPT* time series is affected by various PSN model setups, i.e. treatments. This work will be an extension of our work started in [15]. Additionally, we plan to continue our study of packet traffic fluctuations as a function of network connection topology, routing and incoming traffic volumes that we commenced in [16], [17] and [18].

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